

Predicting Stroke Risk Using Patient Health and Lifestyle Data

Data Analysis Report

Team CodeNexus

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1 Team Details

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2 Abstract

This report presents a stroke-risk prediction study built on patient health and lifestyle data. The original dataset contained 9,722 records, but deduplication revealed 4,612 exact clinical duplicates that had artificially inflated the stroke-positive class. After correction, the true dataset contains 5,110 unique patients with a stroke prevalence of 4.87%.

The analysis combines exploratory data analysis, statistical testing, feature engineering, and leakage-safe machine learning. We evaluate Logistic Regression, Random Forest, XGBoost, Soft Voting, and a Stacking Ensemble. The final results show that threshold tuning is essential in this highly imbalanced clinical setting, and that a simpler model can outperform a more complex one on recall, while stacking achieves the best overall precision-recall tradeoff.

3 1. Introduction

Stroke is a major global health problem and a leading cause of death and long-term disability. The World Health Organization estimates that about 15 million people experience a stroke each year, with roughly 5 million deaths and another 5 million people left permanently disabled. Stroke can cause paralysis, speech impairment, cognitive dysfunction, and vision loss, making early risk detection clinically important.

This study addresses three goals:

- build a machine learning pipeline that remains reliable under severe class imbalance;
- identify the most influential clinical and lifestyle risk factors;
- provide evidence-based insight to support prevention and early screening.

The central methodological contribution is the detection and removal of duplicate stroke-positive records. Training on the original balanced-looking file would produce overly optimistic metrics that do not reflect real-world deployment.

4 2. Dataset and Preprocessing

4.1 2.1 Dataset Description

The dataset contains demographic, medical, and lifestyle variables, with **stroke_event** as the binary target. The main predictors include age, hypertension, heart disease, glucose level, BMI, smoking habit, marital status, employment type, and residence.

Variable	Type	Description	Modelling Status
patient_id	Identifier	Unique record identifier	Excluded
gender	Categorical	Patient gender	Encoded
age	Continuous	Age in years	Retained
has_hypertension	Binary	Hypertension indicator	Retained
has_heart_disease	Binary	Heart disease indicator	Retained
marital_status	Binary	Married or not	Retained
employment_type	Categorical	Work status	Encoded
residence	Categorical	Urban or rural	Encoded
glucose_level	Continuous	Blood glucose level	Retained
bmi_value	Continuous	Body Mass Index	Imputed
smoking_habit	Categorical	Smoking status	Encoded
stroke_event	Binary target	Stroke occurrence	Target

4.2 2.2 Data Integrity

The original file contains 9,722 rows. After removing duplicates using all clinical fields except **patient_id**, the dataset reduces to 5,110 unique patients. This correction is critical because the duplicates were concentrated in the stroke-positive class. Keeping them would leak repeated samples into both training and test partitions and inflate measured performance.

4.3 2.3 Feature Engineering

The final pipeline uses clinically meaningful engineered features:

Feature	Formula	Rationale
age_squared	age^2	Non-linear risk acceleration with age
is_senior	1 if age >= 55	High-risk age flag
age_x_hypertension	age * hypertension	Joint age and blood pressure burden
age_x_heart_disease	age * heart_disease	Combined age and cardiac risk
cvd_count	hypertension + heart_disease	Total cardiovascular burden
glucose_x_bmi	glucose * bmi	Metabolic interaction
glucose_per_bmi	glucose / (bmi + 1)	Risk ratio style feature
bmi_deviation	bmi - 25	Distance from healthy BMI
log_glucose	log(glucose + 1)	Skew reduction
age_over_10	age / 10	Scale adjustment
age_x_glucose	age * glucose	Age-metabolic interaction

4.4 2.4 Missing Value Imputation

BMI missingness is handled using grouped median imputation by gender and age band. This is more realistic than a global median because BMI naturally varies across age and sex groups. Imputation statistics are fit only on training data, then applied separately to validation and test data to avoid leakage.

4.5 2.5 Leakage Prevention

We exclude derived or potentially circular variables from model input:

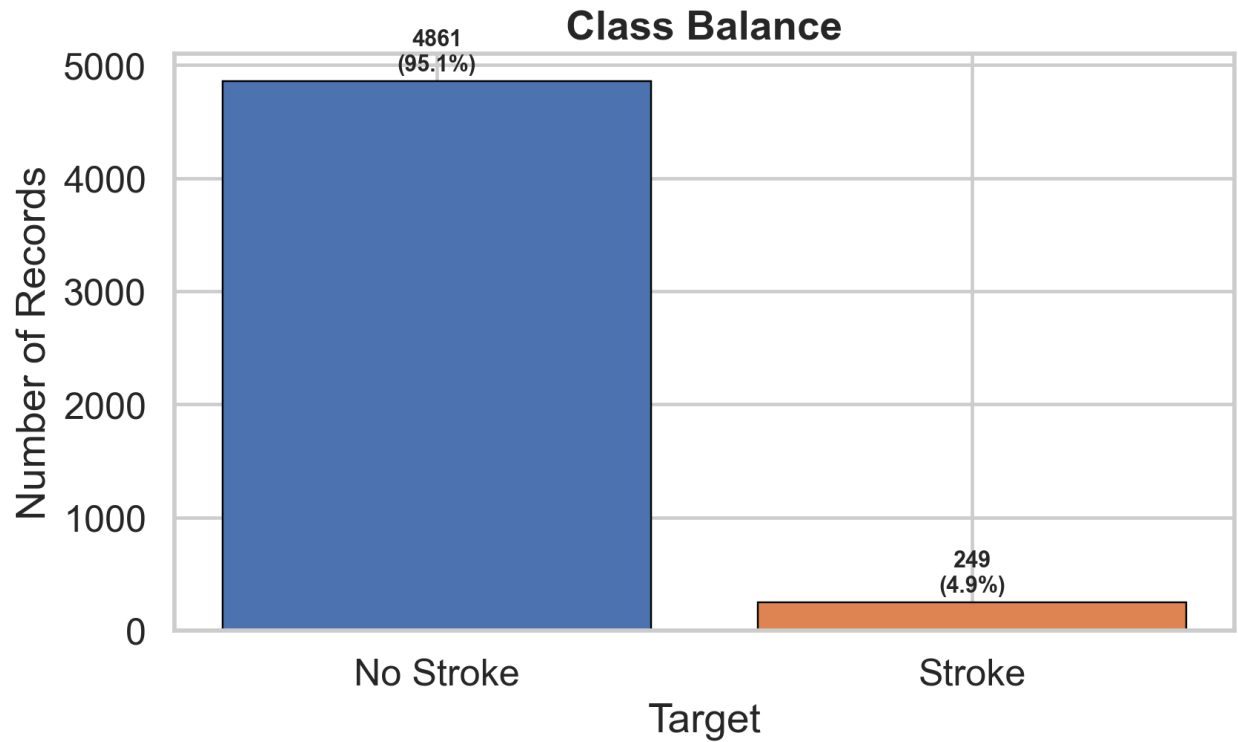
- age_group
- bmi_category
- high_glucose
- risk_score
- lifestyle_risk

These columns are either redundant or too close to the target to be safe for honest prediction.

5 3. Exploratory Data Analysis

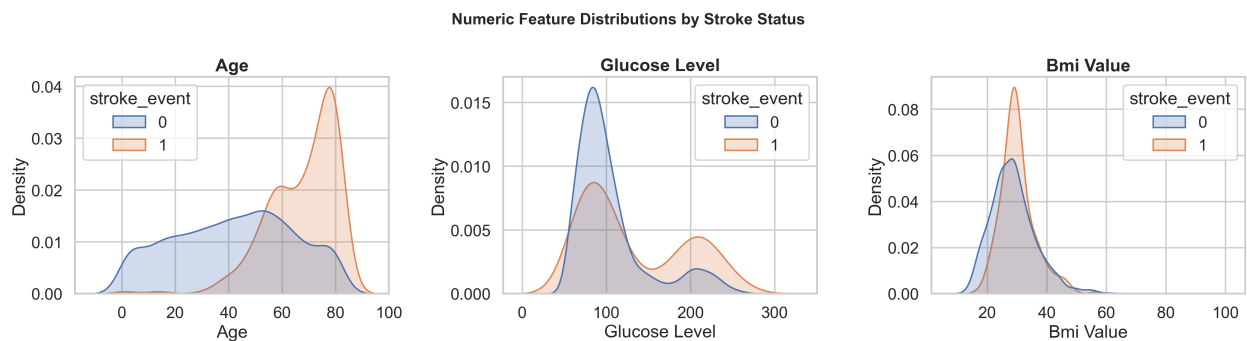
5.1 3.1 Class Balance

The corrected dataset shows the true clinical imbalance: 4,861 non-stroke cases and 249 stroke cases.



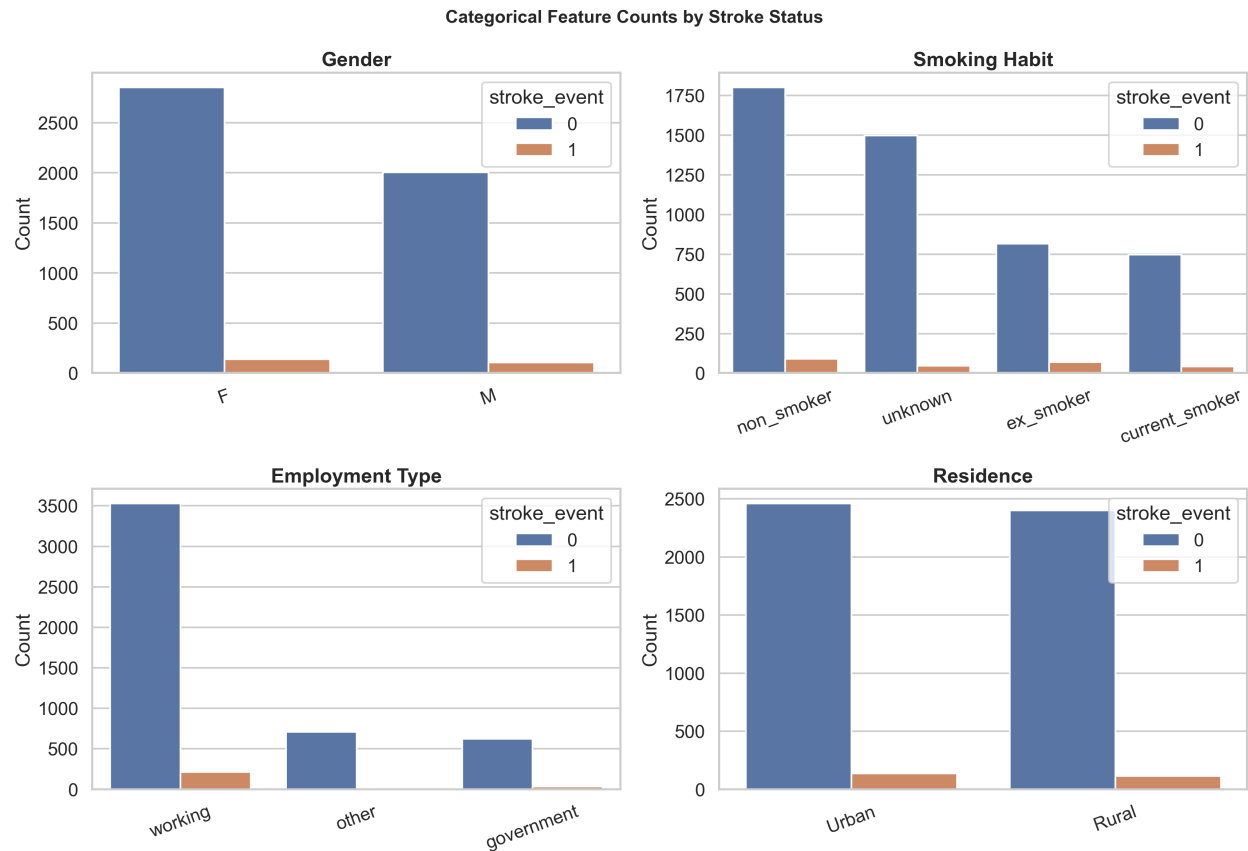
5.2 3.2 Numeric Feature Distributions

Age, glucose level, and BMI all show visible differences between stroke and non-stroke patients. Age is the clearest separator, while glucose and BMI exhibit right-skewed distributions that justify transformation and interaction features.



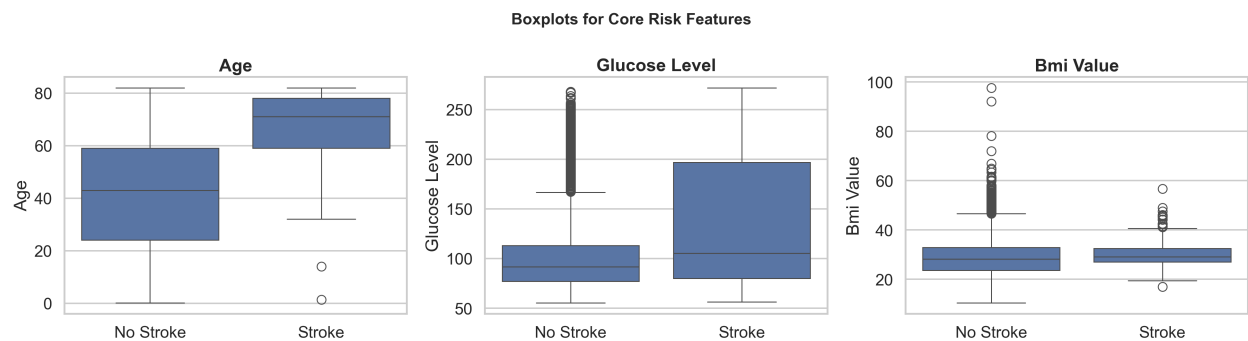
5.3 3.3 Categorical Patterns

Stroke prevalence varies across groups for smoking, employment, residence, and gender. These plots help identify which variables may matter mainly through interaction effects rather than direct linear association.



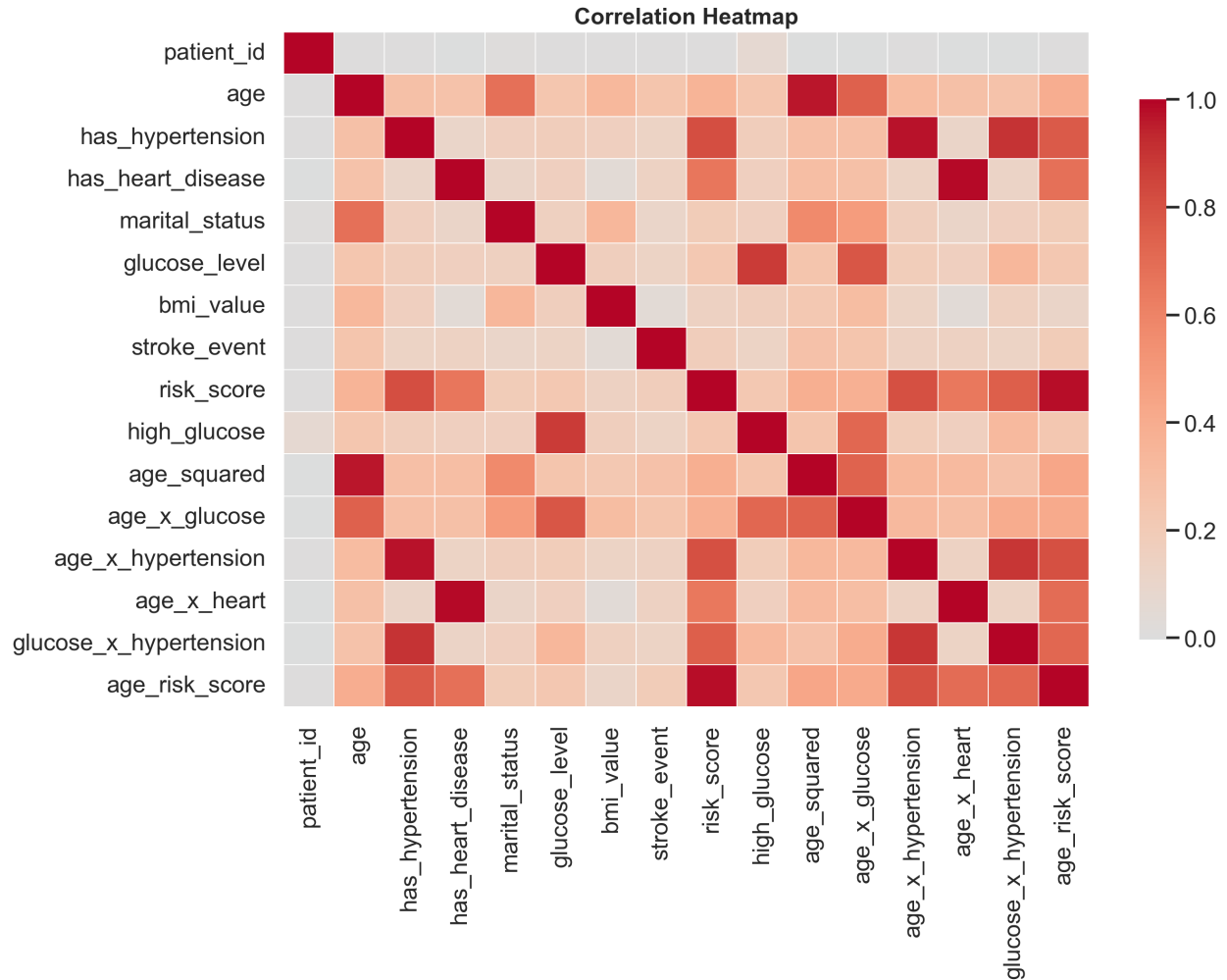
5.4 3.4 Boxplots

Boxplots highlight the shift in age and glucose levels for the stroke class. BMI is more overlapping, but still contributes when combined with other features.



5.5 3.5 Correlation Heatmap

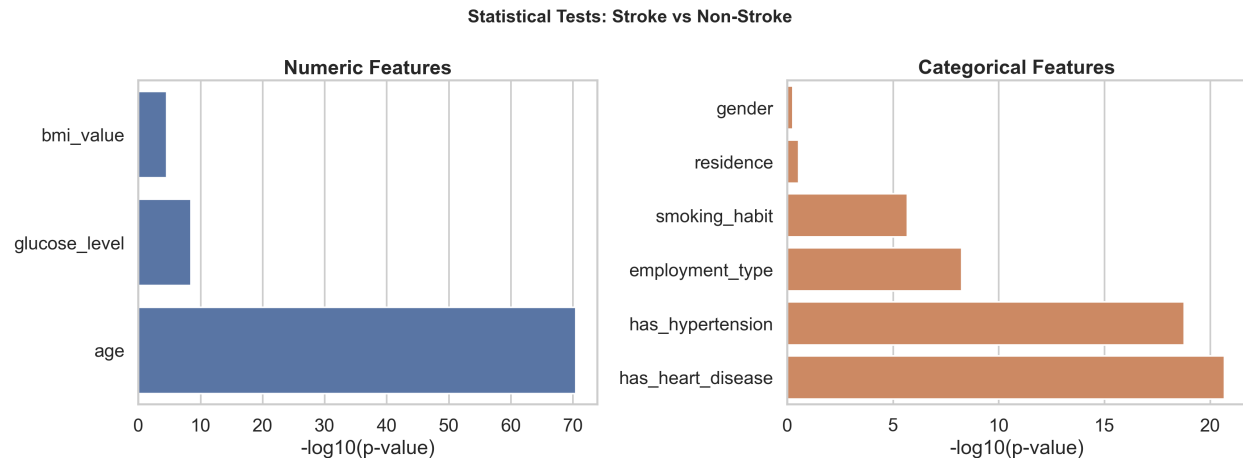
The correlation matrix confirms that age is the strongest single numeric predictor, while the engineered features capture additional non-linear structure.



6 4. Statistical Analysis

6.1 4.1 Hypothesis Testing

We use Mann-Whitney U tests for continuous variables and chi-square tests for categorical variables. This is appropriate because the distributions are not assumed to be normal and several predictors are categorical.



The broad conclusion from these tests is:

- age is significantly different between stroke and non-stroke groups;
- glucose level is significantly higher in the stroke group;
- BMI also differs meaningfully;
- hypertension, heart disease, marital status, and smoking habit show significant categorical association;
- gender and residence are not strongly significant on their own, even though they can still matter in interaction-based models.

6.2 4.2 Interpretation

The statistical tests support the feature engineering strategy. Age is not only the strongest raw predictor, but also benefits from non-linear transformation. Cardiovascular burden works better as a combined indicator than as isolated binary flags.

7 5. Modeling Approach

7.1 5.1 Model Set

Five prediction strategies are evaluated:

- Logistic Regression
- Random Forest
- XGBoost
- Soft Voting
- Stacking Ensemble

7.2 5.2 Imbalance Handling

The class ratio is about 19.5:1, so the pipeline uses three complementary strategies:

- SMOTE on training folds only;
- `class_weight="balanced"` for Logistic Regression and Random Forest;
- `scale_pos_weight` for XGBoost.

7.3 5.3 Stacking Design

Stacking uses out-of-fold predictions to build meta-features. This prevents the meta-learner from seeing labels that already influenced the base predictions.

7.4 5.4 Threshold Tuning

Instead of using the default 0.50 cutoff, thresholds are tuned on a validation set. Two operating modes are defined:

- High Sensitivity mode: prioritize recall with a minimum precision floor of 0.08;
- Balanced mode: maximize F1 while keeping precision at least 0.10.

8 6. Final Results

8.1 6.1 Performance Heatmap

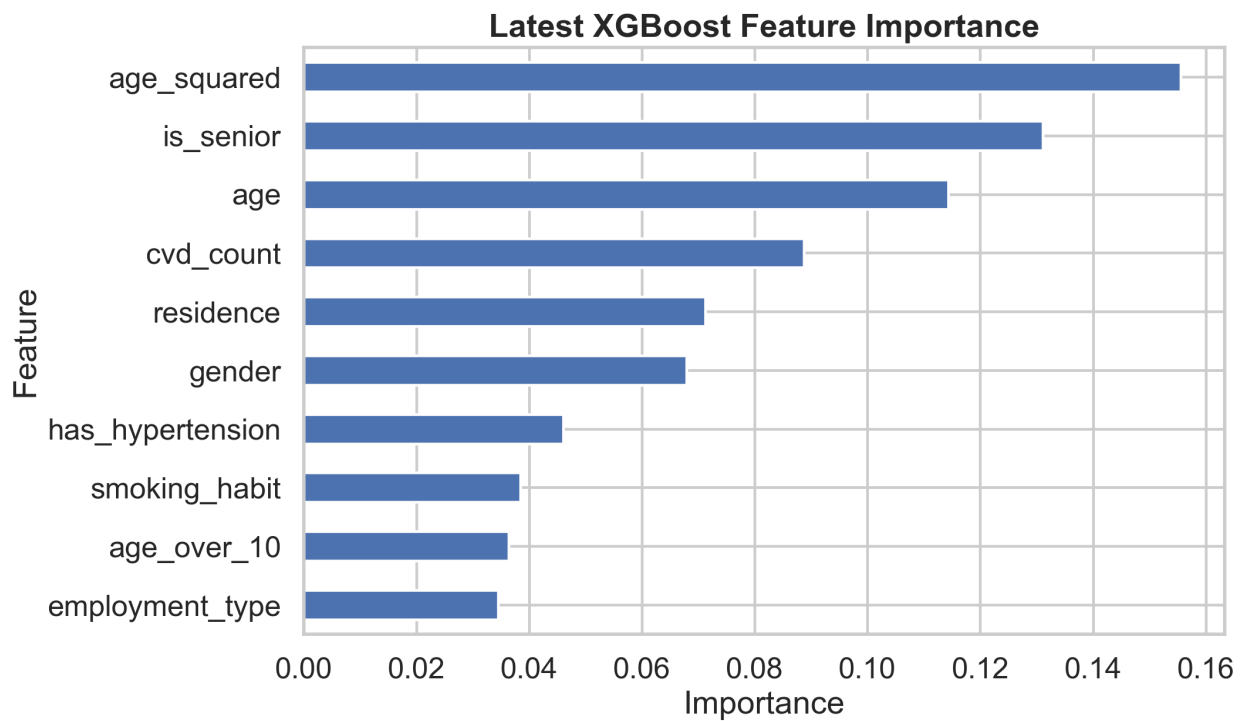
The following summary visualizes the latest model results from the completed run.

Latest Ensemble Performance Summary													
Metric	High Sensitivity					Metric	Balanced						
	AUC-ROC	0.833	0.818	0.803	0.828		0.838	0.833	0.818	0.803	0.828	0.838	
	AUC-PR	0.210	0.183	0.168	0.191		0.215	0.210	0.183	0.168	0.191	0.215	
	Recall	0.940	0.880	0.460	0.820		0.920	0.420	0.420	0.040	0.260	0.780	
	Precision	0.079	0.084	0.168	0.106		0.082	0.223	0.219	0.154	0.200	0.194	
	F1	0.146	0.154	0.246	0.188		0.150	0.292	0.288	0.064	0.226	0.311	
		Logistic Regression	Random Forest	XGBoost	Soft Voting	Stacking			Logistic Regression	Random Forest	XGBoost	Soft Voting	Stacking
		Model					Model						

8.2 6.2 Latest Ensemble Summary

Mode	Model	AUC-ROC	AUC-PR	Recall	Precision	F1
High Sensitivity	Logistic Regression	0.8326	0.2102	0.9400	0.0793	0.1462
High Sensitivity	Random Forest	0.8183	0.1834	0.8800	0.0841	0.1536
High Sensitivity	XGBoost	0.8034	0.1681	0.4600	0.1679	0.2460
High Sensitivity	Soft Voting	0.8282	0.1905	0.8200	0.1059	0.1876
High Sensitivity	Stacking	0.8383	0.2150	0.9200	0.0819	0.1503
Balanced	Logistic Regression	0.8326	0.2102	0.4200	0.2234	0.2917
Balanced	Random Forest	0.8183	0.1834	0.4200	0.2188	0.2877
Balanced	XGBoost	0.8034	0.1681	0.0400	0.1538	0.0635
Balanced	Soft Voting	0.8282	0.1905	0.2600	0.2000	0.2261
Balanced	Stacking	0.8383	0.2150	0.7800	0.1940	0.3108

8.3 6.3 XGBoost Feature Importance



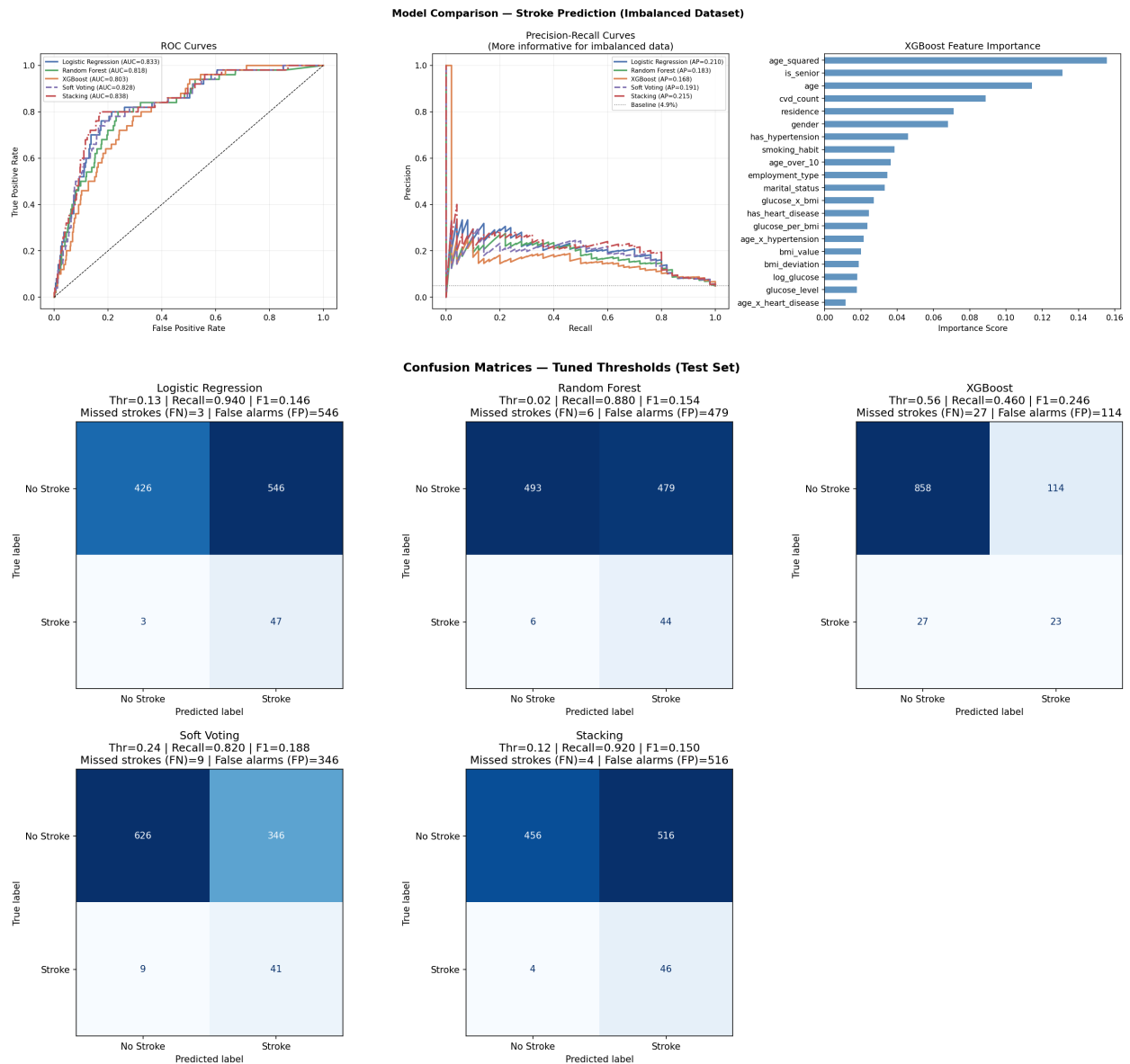
The feature ranking is led by engineered and non-linear terms:

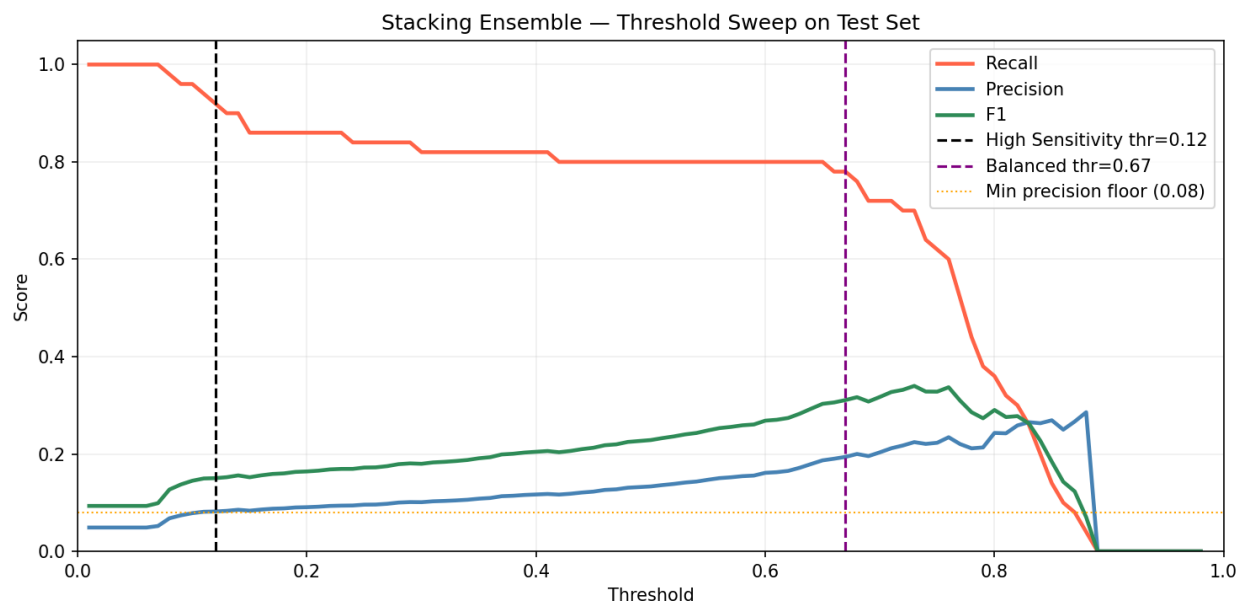
- age_squared
- is_senior
- age
- cvd_count
- residence
- gender

This supports the idea that the strongest signal comes from age and cardiovascular burden, with non-linear structure playing a major role.

8.4 6.4 Ensemble Figures from the Final Model

These plots come directly from `final_stroke_ensemble.py` and summarize the final selected models under the tuned thresholds.





8.5 6.5 Key Performance Interpretation

The results show that:

- Logistic Regression achieves the highest recall in High Sensitivity mode at 0.94;
- Stacking gives the best overall AUC-PR at 0.215;
- XGBoost alone is not the strongest recall model in this small minority-class regime;
- threshold tuning materially changes the operational behavior of every model;
- the Stacking Ensemble gives the best balance between recall and precision in the final comparison.

9 7. Discussion

9.1 7.1 Why Simpler Models Can Win on Recall

The most striking result is that Logistic Regression outperforms XGBoost on recall in the final setting. This is reasonable when the minority class is very small. A simpler regularized linear boundary can generalize better than a more flexible boosting model that has limited minority samples to learn from.

9.2 7.2 Clinical Interpretation

The strongest and most clinically useful findings are consistent with medical knowledge:

- age is the dominant non-modifiable risk factor;
- hypertension and heart disease compound risk;
- glucose and BMI matter, especially through interaction terms;

- smoking status adds useful signal, particularly where data quality is mixed.

9.3 7.3 Deployment Considerations

For screening use, a high-recall model is often more useful than one with high precision. In this case, a two-stage protocol is sensible: the model flags high-risk patients, then clinicians review the flagged group.

10 8. Limitations

- The dataset comes from a single source and needs external validation.
- The minority class remains small even after deduplication.
- The data is cross-sectional rather than longitudinal.
- Smoking status includes a large unknown category.
- Any model performance should be interpreted in light of the original duplication issue.

11 9. Conclusion

This project develops a leakage-safe stroke prediction workflow on a genuinely imbalanced dataset. After removing artificial duplicates, the true prevalence is 4.87%, and the modeling pipeline is adapted accordingly using SMOTE, class weighting, threshold tuning, and stacking.

The final results show that threshold selection is crucial for clinical utility. Logistic Regression gives the highest recall in high-sensitivity operation, while Stacking delivers the best overall precision-recall performance. The visual analysis and statistical testing reinforce the same conclusion: age and cardiovascular burden dominate stroke risk in this dataset.

12 10. Recommendations

- Use the model as a screening aid, not a diagnostic replacement.
- Prioritize patients with elevated age, hypertension, and heart disease.
- Monitor glucose and BMI carefully in older adults.
- Validate the model on a second real-world cohort before deployment.
- Extend future work with SHAP explanations and probability calibration.

13 11. Future Work

- External validation on a new patient cohort.
- SHAP-based interpretability analysis.
- Probability calibration for better risk communication.
- Longitudinal modeling with time-aware patient trajectories.

14 References

- [1] World Health Organization, “Stroke, Cerebrovascular accident,” WHO, 2023.
- [2] Centers for Disease Control and Prevention, “Stroke Facts and Statistics,” CDC, 2023.
- [3] Global Burden of Disease Study, “Global, regional, and national burden of stroke,” The Lancet Neurology, 2021.
- [4] S. K. Pandey et al., “Machine learning techniques for stroke prediction: A review,” IEEE Access, 2020.

15 Appendix: Contribution Statement

Member Name	Primary Contributions
R.P.M. Vithanage	Data cleaning, deduplication, BMI imputation, EDA
S.E.N. S. Silva	Statistical testing, feature engineering, visualisation